RISE: Supplementary Experimental Analyses

Research supplement to the supplied manuscript

23 September 2026

This supplement reports completed exploratory analyses and matched controls. RISE is named SISA in the implementation; AM and PA correspond to decay and rotation. Native A denotes an independently trained backbone, whereas B retains auxiliary content projections. C, D, and E denote AM-only, PA-only, and Full RISE. The original 16-cell, seed-20262027 main table is unchanged. New controls use the matched seed-20262028 protocol, not replacements selected for the original table. Intervals quantify test-user sampling uncertainty conditional on fitted models, not training-seed variation. The accompanying Chinese dossier contains the full execution ledger, action-specific subgroup support, rejected diagnostics and reproduction records. These materials are proposed manuscript additions; the supplied manuscript PDF has not been edited.

1 Sample-conditional effects and statistical validation

1.1 Conditional benefits relative to the native backbone

A distinct comparison. We evaluate Full RISE (E) against a newly completed native HyFormer control (A) under the matched MerRec seed-20262028 protocol. The native model was trained without the auxiliary RISE branch; Content-only (B), used in the companion component analysis, retains auxiliary content projections. Accordingly, $\delta_g^{E:A} = A_{E,g} - A_{A,g}$ measures the full intervention, whereas $\delta_g^{E:B}$ measures the additional difference between separately trained modulation configurations. These estimands should not be substituted for one another. Both comparisons use the same Full RISE predictions and 11,599,872 retained test rows. All prediction-shard checksums, physical sample identities, six metadata fields, and five labels were verified. Evaluation follows the original eight-block incomplete-batch removal, rather than silently adding the omitted test tails.

The unrounded prediction-based macro ROC-AUC increases from 0.785633637 to 0.800872336, a difference of 0.015238700 with a nominal user-cluster interval $[0.013878878, 0.016598521]$ (Table 1). All five actions improve globally, but this does not imply that every subgroup or every quality metric improves. The comparison concerns this new matched seed, not recovery of the unavailable original seed-20262027 model instance.

Sample conditions. Table 2 retains all one-dimensional groups. Their boundaries were fixed before formal effect inspection, with frequency cut points derived from training counts. Effective length excludes the target and counts valid positions in a 100-token window. Frequency counts use training targets and global IDs; positive-frequency entity quartiles are 14/80 for users and 1/4 for items. Zero frequency means unseen as a training target, without establishing strict cold start. At fixed length 100, category entropy H and candidate-category history fraction Q exclude category codes 0 and 309; undefined values remain separate. Row counts are shared across actions, while positive counts remain label-specific and are never pooled into a macro count.

The macro gain is 0.019479 at length 100 and 0.012081 at lengths 1–20. A direct contrast gives 0.007398, with interval $[0.003664, 0.011133]$. Empty history instead yields 0.005278, with interval $[-0.004541, 0.015097]$ crossing zero. The positive Content-only-controlled empty-history result therefore cannot be reused as native-backbone evidence. Candidate and context inputs remain available, and separate training and native padding behavior prevent a direct attribution to history accumulation. The 51–99 point estimate also exceeds that at 100, so the pattern is not strictly monotone.

Frequency effects are positive in every reported macro group but are not monotone. Item-frequency gains are 0.011887, 0.016704, 0.017181, and 0.015555 for zero, one, two–four, and at least five occurrences. User-frequency gains are 0.014254, 0.014794, 0.019759, and 0.019177. These marginal comparisons do not separate frequency from history length, user composition, or label difficulty.

Fixed-length diversity is likewise nonmonotone: gains across the four defined entropy groups are 0.004035, 0.022411, 0.014523, and 0.021134. Although the high-versus-middle entropy contrast is positive, 0.006610 $[0.000914, 0.012306]$, this local contrast does not imply increasing gains across the full entropy range. For sparse nonzero versus high

Table 1: Global native-backbone comparison on the same 11,599,872 retained rows. AUC differences use unrounded predictions. Intervals are nominal 95% user-cluster intervals; macro AUC equally weights five actions and has no pooled positive count.

Action	Positives	Native (A)	Full RISE (E)	Δ	Nominal 95% interval
Like	1,432,490	0.729441	0.750689	+0.021248	[+0.020882, +0.021615]
Cart	128,634	0.787154	0.810189	+0.023034	[+0.022272, +0.023797]
Offer	39,437	0.726794	0.731796	+0.005002	[+0.003269, +0.006735]
Checkout	7,651	0.842007	0.855579	+0.013572	[+0.010433, +0.016711]
Purchase	4,642	0.842773	0.856109	+0.013337	[+0.009363, +0.017310]
Macro	–	0.785634	0.800872	+0.015239	[+0.013879, +0.016599]

candidate-category support, the effect difference is only 0.001487 $[-0.006770, 0.009744]$. Thus, the corresponding Content-only-controlled contrast does not establish the same claim against the native backbone. The three contrasts were selected after inspecting the initial grouped mean effects of Full RISE minus Content-only; this native analysis reuses those same definitions. They remain exploratory, not preregistered or confirmatory, with no multiplicity correction.

Task and probability-quality exceptions. The complete subgroup inventory has four negative task-level ROC-AUC point estimates: Offer and Checkout at empty history, and the same actions in the near-zero-entropy length-100 group. Every corresponding nominal interval crosses zero. The latter group has only 24 Offer and four Checkout positives; it also has just two Purchase positives. Undefined candidate support has no Checkout or Purchase positives, so its five-action macro AUC is unavailable rather than recomputed over fewer actions.

We retain original probabilities and supplement ROC-AUC with non-interpolated average precision (AP), Brier score, and Logloss (Table 4). Macro AP increases from 0.077907499 to 0.092808705, macro Brier decreases from 0.022998024 to 0.022297116, and macro Logloss decreases from 0.085562731 to 0.082677531. Nevertheless, Purchase AP decreases from 0.012851605 to 0.012773943 despite its ROC-AUC improvement. Offer and Checkout each have small increases in both Brier and Logloss. These descriptive differences show that ranking gains do not guarantee uniform probability-quality gains; they have no new confidence intervals. No calibrator was fitted and no decision threshold was selected. Fixed-grid calibration error is bin-dependent and is not used as evidence of uniformly improved calibration.

AUC structure and inferential scope. For each action, the exact identity

$$A_m = \sum_{g,h:n_g^+ n_h^- > 0} \frac{n_g^+ n_h^-}{N^+ N^-} A_m^{g,h} \tag{1}$$

separates positive–negative pairs within and across the five length groups, with half credit for ties. Equal-label averaging decomposes the native-control macro gain into 0.007672426 within-group and 0.007566274 cross-group contributions; the latter accounts for 49.65%, not the component comparison's 51.76%. The maximum reconstruction residual is 1.74×10^{-16}. This is an algebraic allocation for the chosen partition, not causal attribution.

Intervals aggregate normalized paired AUC influences by user. Group contrasts retain shared-user covariance, and macro intervals average the five label influences before estimating variance. They condition on fixed trained models and do not include seed variability. A user's contribution combines all of that user's retained observations, including positive and negative examples. The variance includes the finite-cluster correction; actions sharing users are not counted as independent replications. The earlier exact-bootstrap check concerned E versus Content-only; it was not relabeled as an empirical bootstrap validation of E versus native A. One seed, added model capacity, correlated grouping variables, HyFormer padding, and unresolved availability times of MerRec's attributed historical outcomes limit causal and deployment interpretations. The native comparison should establish the primary conditional performance result; the component analysis then addresses the separate incremental AM/PA question.

Table 2: All input-defined groups, with native A as the leftmost model column. AUC is the equal mean over five actions; $\Delta = E - A$ in absolute AUC units. All intervals are nominal and conditional on the fixed models.

Group	N	Native (A)	Full RISE (E)	Δ	Nominal 95% interval
Effective history length L					
0	14,208	0.583720	0.588998	+0.005278	[-0.004541, +0.015097]
1–20	852,540	0.709818	0.721899	+0.012081	[+0.009411, +0.014751]
21–50	1,157,429	0.715041	0.731961	+0.016920	[+0.012155, +0.021684]
51–99	1,394,992	0.735010	0.754913	+0.019904	[+0.014615, +0.025192]
100	8,180,703	0.777188	0.796667	+0.019479	[+0.016868, +0.022090]
Item training-target frequency					
0	3,506,086	0.788862	0.800749	+0.011887	[+0.009641, +0.014132]
1	1,173,599	0.798687	0.815392	+0.016704	[+0.011850, +0.021558]
2–4	2,277,671	0.790684	0.807865	+0.017181	[+0.013931, +0.020431]
≥ 5	4,642,516	0.789655	0.805211	+0.015555	[+0.013458, +0.017652]
User training-target frequency					
0	523,621	0.783404	0.797658	+0.014254	[+0.011778, +0.016729]
1–14	967,941	0.723387	0.738182	+0.014794	[+0.011096, +0.018492]
15–80	2,444,476	0.737312	0.757071	+0.019759	[+0.015996, +0.023522]
≥ 81	7,663,834	0.780522	0.799699	+0.019177	[+0.016491, +0.021862]
Category entropy H, L = 100					
Undefined	0	–	–	–	–
$H \leq 10^{-7}$	22,254	0.872793	0.876828	+0.004035	[-0.007299, +0.015368]
$10^{-7} < H \leq \ln 2$	574,067	0.791962	0.814373	+0.022411	[+0.014014, +0.030808]
$\ln 2 < H \leq \ln 5$	1,981,517	0.787908	0.802431	+0.014523	[+0.009949, +0.019098]
$H > \ln 5$	5,602,865	0.770286	0.791419	+0.021134	[+0.017729, +0.024538]
Candidate-category fraction Q, L = 100					
Undefined	1,431	–	–	–	–
0	932,342	0.768041	0.791313	+0.023272	[+0.015393, +0.031151]
$0 < Q \leq 0.25$	4,232,607	0.767981	0.788132	+0.020152	[+0.016149, +0.024155]
$0.25 < Q \leq 0.75$	2,243,066	0.789502	0.806832	+0.017330	[+0.013264, +0.021396]
$Q > 0.75$	771,257	0.792607	0.811271	+0.018665	[+0.011430, +0.025899]

Undefined entropy has zero rows. Undefined candidate support has no Checkout or Purchase positives, so five-action macro AUC and its interval are unavailable. The near-zero-entropy group has only four Checkout and two Purchase positives; its displayed estimates and interval should be read descriptively.

Table 3: Inherited exploratory native-control contrasts of five-action macro $E - A$ effects. Definitions were selected after inspecting the initial grouped mean Full RISE-minus-Content-only effects and reused for this native analysis. Entropy and candidate-support contrasts restrict length to 100. Shared-user and across-label covariance are retained; no multiplicity correction or training-seed uncertainty is represented.

First group minus second group	Effect difference	Nominal 95% interval
Length: 100 minus 1–20	+0.007398	[+0.003664, +0.011133]
$H > \ln 5$ minus $\ln 2 < H \leq \ln 5$	+0.006610	[+0.000914, +0.012306]
$0 < Q \leq 0.25$ minus $Q > 0.75$	+0.001487	[-0.006770, +0.009744]

Table 4: Probability-quality checks on identical paired rows. AP is non-interpolated average precision, not trapezoidal PR-AUC. All differences are Full RISE minus native A; higher AP and lower Brier/Logloss are preferable. These are descriptive point estimates without new uncertainty intervals.

Action	Native AP	Full RISE AP	ΔAP	ΔBrier	ΔLogloss
Like	0.290451	0.341571	$+5.112 \times 10^{-2}$	-3.336×10^{-3}	-1.242×10^{-2}
Cart	0.056999	0.076324	$+1.932 \times 10^{-2}$	-1.680×10^{-4}	-1.999×10^{-3}
Offer	0.012898	0.013471	$+5.729 \times 10^{-4}$	$+1.619 \times 10^{-6}$	$+1.513 \times 10^{-5}$
Checkout	0.016337	0.019904	$+3.567 \times 10^{-3}$	$+6.093 \times 10^{-7}$	$+1.518 \times 10^{-5}$
Purchase	0.012852	0.012774	-7.766×10^{-5}	-2.931×10^{-6}	-3.970×10^{-5}
Macro	0.077907	0.092809	$+1.490 \times 10^{-2}$	-7.009×10^{-4}	-2.885×10^{-3}

1.2 When does cumulative modulation help?

Protocol and estimands. We extend the matched HyFormer–MerRec component study using the existing Content-only, AM-only, PA-only, and Full RISE checkpoints at seed 20262028. The principal comparison is Full RISE versus Content-only, which retains the auxiliary content projections; it therefore measures the additional effect of the trained modulation configuration, rather than the benefit of introducing the entire auxiliary branch into the unaugmented backbone. All configurations use identical labels and 11,599,872 retained test rows across eight physical blocks, following the original block-level incomplete-batch removal. Prediction shard checksums, sample identities, and five-label equality were verified. The resulting global macro AUC is 0.783121081 for Content-only and 0.800872336 for Full RISE.

Effective history length L counts valid, non-target positions in the last 100-token window. Training frequencies count target occurrences using global entity identifiers, with zero separated from observed entities. Positive-frequency entity quartiles are $(14, 80)$ for users and $(1, 4)$ for items; these are entity, not interaction, quantiles. At $L = 100$, we measure category entropy $H = -\sum_c p_c \log p_c$ and the fraction Q of known history categories matching the candidate. Definite unknown code 0 and ambiguous code 309 are excluded; the latter affects only 24 history positions and no target in this population. Undefined fractions remain separate. Group boundaries were fixed before formal effect inspection, with frequency cut points derived from training counts. This boundary choice is distinct from selecting group contrasts after inspecting the initial results. Table 5 reports all one-dimensional groups. N always counts shared examples; positive counts and prevalence remain action-specific and are never pooled across actions. Macro AUC equally weights the same five actions.

For a group g, let $\delta_g = A_{\text{Full},g} - A_{\text{Content},g}$. We aggregate normalized paired AUC influences within users. If z_{ug} is user u's contribution, the variance estimator is

$$\widehat{\text{Var}}(\delta_g) = \frac{K_g}{K_g - 1} \sum_{u=1}^{K_g} (z_{ug} - \bar{z}_g)^2. \tag{2}$$

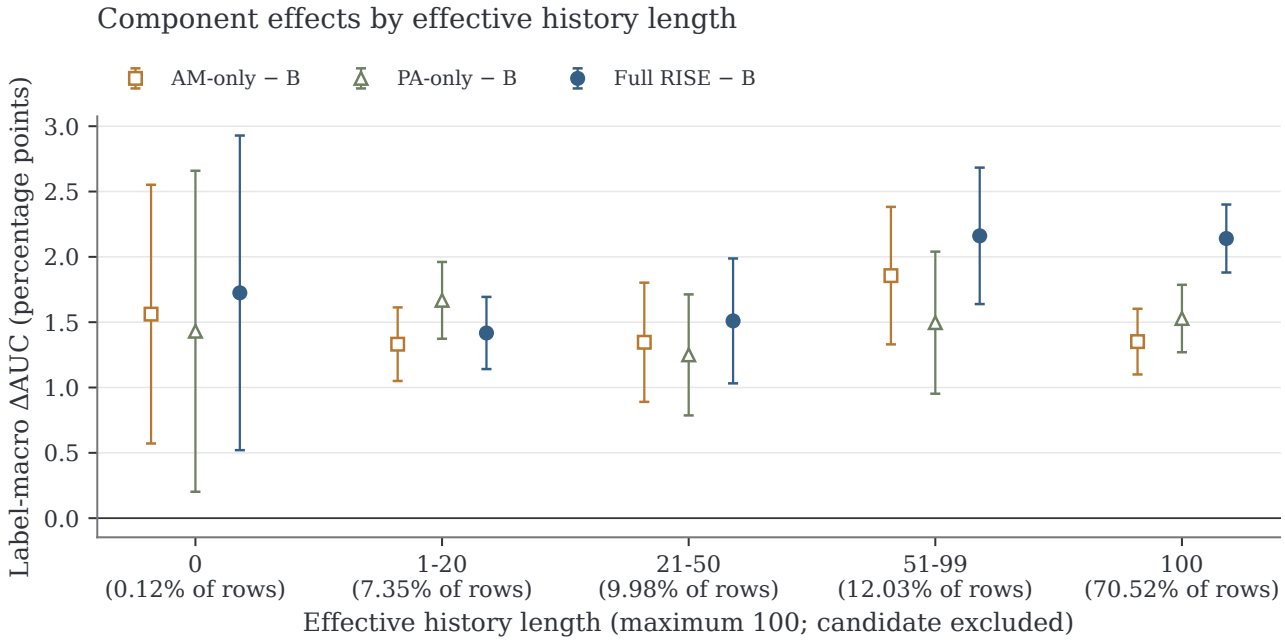
Group contrasts use $z_{ug} - z_{uh}$ on the union of users; macro contrasts average label influences before estimating variance. This retains shared-user and shared-label covariance. Intervals are nominal 95% normal intervals, conditional on the trained models, without multiplicity correction. The three supplemental contrasts were explicitly selected after inspecting the initial grouped mean effects of Full RISE minus Content-only. They are exploratory, not preregistered or confirmatory comparisons.

Conditional performance. Full RISE improves macro AUC across all five length groups, but the effect is not monotone in length (Figure 1). The gain is 0.014173 for $L = 1$–20 and 0.021404 at $L = 100$; their difference is 0.007231 with interval $[0.003436, 0.011026]$. The $L = 51$–99 estimate is slightly larger than that at 100. The empty-history group also improves by 0.017246 ($N = 14,208$). This does not measure accumulation over an absent history: separately trained configurations can learn different shared representations and decoder parameters, while candidate and context inputs remain available. The empty-history Offer point estimate is nevertheless negative (-0.007558), so the macro result is not a uniform task response. HyFormer's native padding treatment also limits a direct mechanistic reading of length differences.

Frequency groups show that the effect is not simply largest for the coldest entities (Figure 2). Items occurring once as training targets gain 0.024028, compared with 0.011569 for unseen training targets and 0.017723 for frequency at least five. Users with frequency above 80 gain 0.022382, but the user-frequency pattern is likewise not strictly monotone. "Unseen training target" does not establish absence from every training history, and frequency and length remain correlated.

Holding visible length at 100, the gain rises from 0.017190 at $\ln 2 < H \leq \ln 5$ to 0.024195 at $H > \ln 5$. The difference is 0.007005 $[0.001282, 0.012728]$. Sparse but nonzero candidate-category support, $0 < Q \leq 0.25$, yields a larger gain than $Q > 0.75$ by 0.008206 $[0.000438, 0.015974]$. These patterns are compatible with conditional evidence selection, but category entropy and matching are proxies, not measurements of PA's learned vector alignment. They do not isolate category structure from other user or task characteristics.

A task-specific counterexample. Offer exposes a limitation of aggregate component rankings. Full RISE minus PA-only equals -0.021456 at $L = 1$–20. Its interval is $[-0.026858, -0.016054]$. At $L = 100$, the difference is -0.000138, with interval $[-0.002385, 0.002109]$. The explicitly post-hoc difference between these conditional effects is 0.021317 $[0.015468, 0.027167]$, preserving covariance for 932 shared users. Table 6 distinguishes this selected-label analysis from the three five-action contrasts. It identifies heterogeneity between trained configurations, not a causal effect of a particular AM gate or proof that AM is uniformly harmful for Offer.



HyFormer-MerRec, seed 20262028; 11,599,872 paired test rows.
Bars: 95% user-cluster linearized intervals. B = Content-only; it is not the native baseline.

Figure 1: Verified component effects by effective history length. In the figure, B denotes Content-only. Points compare independently trained component configurations; bars are nominal user-cluster intervals, not training-seed intervals. The vertical axis uses AUC percentage points, whereas the tables report absolute AUC differences.

AUC structure and uncertainty checks. Within-group AUCs cannot be averaged using group sample shares to recover the global metric. For each action, the exact decomposition is

$$A_m = \sum_{g,h:\, n_g^+ n_h^- > 0} \frac{n_g^+ n_h^-}{N^+ N^-} A_m^{g,h}, \tag{3}$$

where $A_m^{g,h}$ compares positives from g with negatives from h, assigning half credit to ties. Averaging contributions equally across actions reconstructs the macro gain of 0.017751256: 0.008562778 comes from same-length-group pairs and 0.009188478 from cross-length-group pairs (51.76%). The maximum numerical residual is 1.67×10^{-16}. This is a partition-specific algebraic allocation, not attribution of causal responsibility to samples or components.

We additionally selected 10,000 users uniformly without replacement using a fixed seed, retained all their 165,352 test rows, and performed 500 exact user bootstrap replicates with shared multiplicities across models and actions. For Like, Offer, and Purchase, bootstrap-to-linearized standard-error ratios on this *same subsample* are 1.061, 0.994, and 1.062; all 500 replicates per action contain both classes. Purchase has only 66 positives across 57 users. The comparison supports a local uncertainty check, not a guarantee for every sparse group or an estimate of training-seed variability; no agreement threshold was imposed or unsuccessful draw replaced.

Interpretation and limits. The evidence concerns one backbone, dataset, and component seed. Active dense parameter counts differ across configurations, and local-only or fixed-control comparisons are not supplied by these analyses. Short histories also change padding exposure. MerRec history actions encode outcomes attributed to earlier view anchors; original action timestamps are unavailable here, so point-in-time availability is not established. Rare groups require particular restraint: the near-zero-entropy group has only four Checkout and two Purchase positives, and undefined candidate support has none for these two actions. Its five-action macro AUC is therefore unavailable, rather than recomputed over fewer actions. The results characterize conditional offline behavior under this protocol; they neither establish temporal causality nor explain all gains through cumulative historical evidence.

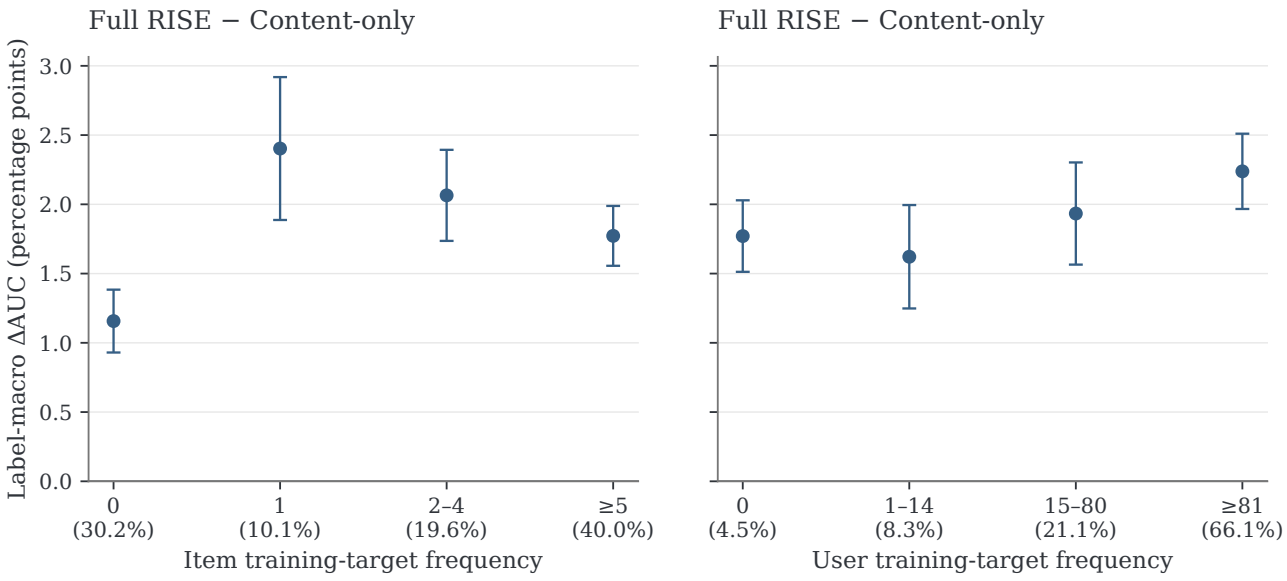
Table 5: All input-defined groups in the verified HyFormer–MerRec component study. AUC is the equal mean of five action-specific AUCs; Δ = Full RISE − Content-only in absolute AUC units. Differences and intervals are computed before display rounding.

Group	N	Content-only	Full RISE	Δ	Nominal 95% interval
Effective history length L					
0	14,208	0.571752	0.588998	+0.017246	[+0.005205, +0.029287]
1–20	852,540	0.707726	0.721899	+0.014173	[+0.011411, +0.016935]
21–50	1,157,429	0.716863	0.731961	+0.015098	[+0.010319, +0.019877]
51–99	1,394,992	0.733304	0.754913	+0.021610	[+0.016388, +0.026831]
100	8,180,703	0.775263	0.796667	+0.021404	[+0.018802, +0.024006]
Item training-target frequency F_i					
0	3,506,086	0.789179	0.800749	+0.011569	[+0.009300, +0.013839]
1	1,173,599	0.791363	0.815392	+0.024028	[+0.018869, +0.029187]
2–4	2,277,671	0.787217	0.807865	+0.020647	[+0.017360, +0.023935]
≥ 5	4,642,516	0.787488	0.805211	+0.017723	[+0.015562, +0.019884]
User training-target frequency F_u					
0	523,621	0.779950	0.797658	+0.017708	[+0.015126, +0.020290]
1–14	967,941	0.721967	0.738182	+0.016215	[+0.012481, +0.019948]
15–80	2,444,476	0.737735	0.757071	+0.019336	[+0.015647, +0.023024]
≥ 81	7,663,834	0.777317	0.799699	+0.022382	[+0.019664, +0.025100]
Category entropy H, restricted to $L = 100$					
Undefined	0	–	–	–	–
$H \leq 10^{-7}$ ‡	22,254	0.865468	0.876828	+0.011360	[−0.001586, +0.024306]
$10^{-7} < H \leq \ln 2$	574,067	0.798711	0.814373	+0.015662	[+0.008194, +0.023129]
$\ln 2 < H \leq \ln 5$	1,981,517	0.785241	0.802431	+0.017190	[+0.012581, +0.021799]
$H > \ln 5$	5,602,865	0.767224	0.791419	+0.024195	[+0.020784, +0.027606]
Candidate-category history fraction Q, restricted to $L = 100$					
Undefined†	1,431	–	–	–	–
0	932,342	0.770007	0.791313	+0.021306	[+0.013450, +0.029163]
$0 < Q \leq 0.25$	4,232,607	0.764804	0.788132	+0.023328	[+0.019287, +0.027370]
$0.25 < Q \leq 0.75$	2,243,066	0.786034	0.806832	+0.020798	[+0.016610, +0.024986]
$Q > 0.75$	771,257	0.796149	0.811271	+0.015123	[+0.008478, +0.021767]

Intervals use paired user-cluster influences with trained models fixed. Frequency zero means unseen among training targets, not proven strict cold start. Entropy groups exclude category codes 0 and 309 and use natural logarithms. †Undefined candidate support has no Checkout or Purchase positives; five-action macro AUC and its interval are unavailable. ‡The near-zero-entropy group has four Checkout and two Purchase positives; its displayed estimates and nominal interval are fragile and should be read descriptively. The undefined-entropy group is empty.

Table 6: Exploratory differences between conditional effects, accounting for users shared across groups. The first three rows concern the five-action Full RISE minus Content-only effect δ and were selected after inspecting the initial grouped mean effects. The final row is the later-added post-hoc Offer comparison of Full RISE (E) and PA-only (D). All comparisons are exploratory. Intervals are nominal and unadjusted for multiplicity.

Contrast	Effect difference	Nominal 95% interval
$\delta_{L=100} - \delta_{L=1\text{–}20}$	+0.007231	[+0.003436, +0.011026]
$\delta_{H>\ln 5} - \delta_{\ln 2<H\leq\ln 5}$ at $L = 100$	+0.007005	[+0.001282, +0.012728]
$\delta_{0<Q\leq 0.25} - \delta_{Q>0.75}$ at $L = 100$	+0.008206	[+0.000438, +0.015974]
Offer: $(E - D)_{L=100} - (E - D)_{L=1\text{–}20}$	+0.021317	[+0.015468, +0.027167]



HyFormer–MerRec; 95% user-cluster linearized intervals. Parentheses: share of retained test rows.
Cut points use positive-frequency training entities (Q25/Q75); 0 means unseen as a training target.

Figure 2: Full RISE minus Content-only by item and user training-target frequency. Cut points use the positive-frequency entity distribution; parentheses show retained sample shares. Marginal frequency comparisons do not separate frequency from history length or other correlated properties.

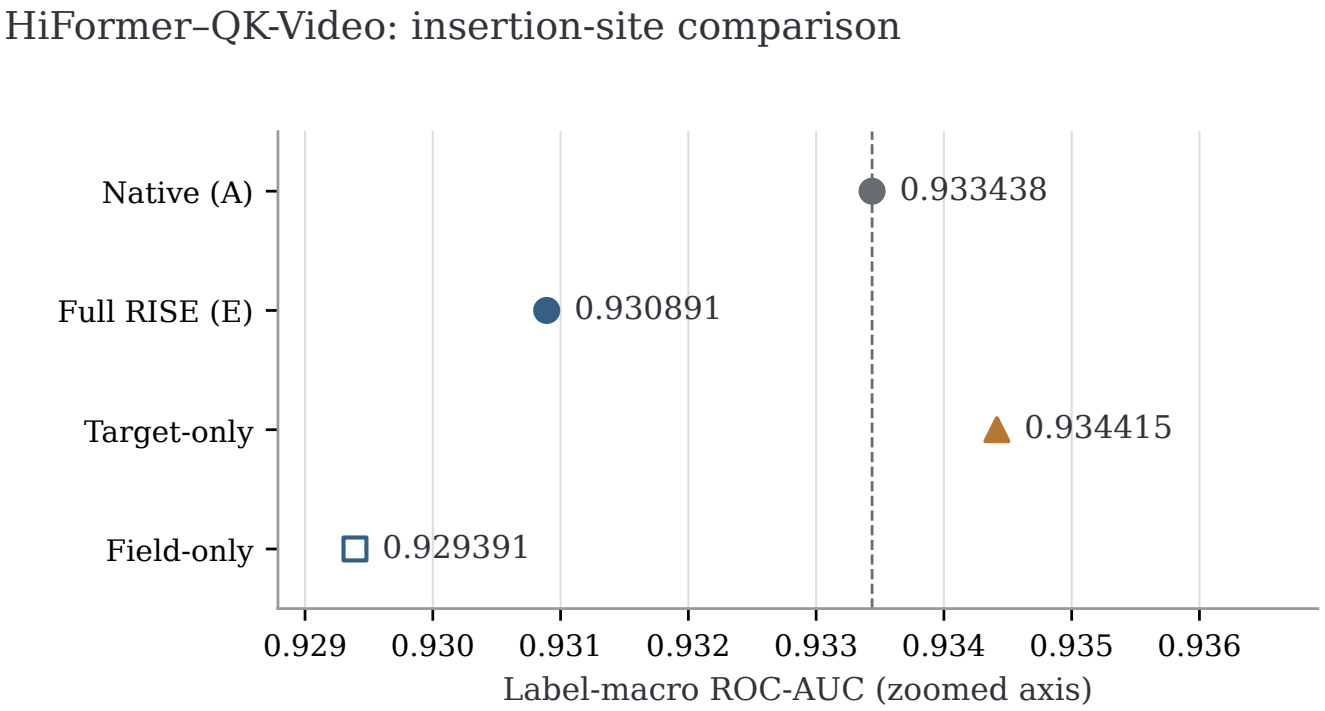


Figure 3: HiFormer–QK-Video placement comparison. Macro AUC equally weights four actions. The displayed AUC axis is zoomed; there are no training-seed uncertainty intervals.

2 Architectural controls, efficiency and mechanism diagnostics

2.1 Placement, input-independent controls, and realized cost

Insertion-site sensitivity. We retain the matched seed-20262028, one-epoch protocol and compare the native backbone, full RISE, and selective-site variants. On HiFormer–QK-Video, macro ROC-AUC is 0.9334380 for native A, 0.9308912 for full RISE, 0.9344147 for target-only insertion, and 0.9293908 for field-only insertion (Figure 3). The target-only variant improves over native A by +0.0009767 and full RISE by +0.0035235 in absolute AUC, whereas the field-only variant is below both. This identifies insertion layout as a plausible contributor to the observed regression; it does not isolate a unique temporal mechanism. Selective-site variants have different auxiliary parameter counts, so this comparison is not capacity-matched. All results are single-seed point estimates. Additional QK-Video placement results are retained in Table 7, rather than selecting only the best-performing placement for each backbone.

Native-controlled history groups. The grouped comparison uses 48,529,408 identical paired QK-Video rows and 153,157,632 TAAC-25 rows. Full-minus-native macro AUC is negative in all four observed QK-Video length groups, with nominal intervals below zero. On TAAC-25, the 1–20 estimate is -0.001685 with interval $[-0.003972, +0.000602]$; the estimates at lengths 21–50 and 51–99 are +0.023980 and +0.016602. Empty-history TAAC-25 has a wide interval containing zero; QK-Video has no empty-history rows, and TAAC-25 has no length-100 rows in the retained test set. Figure 4 retains these empty categories explicitly. The intervals are normal 95% user-cluster linearized intervals conditional on fixed trained models. Macro influences are averaged across actions before variance estimation. They contain neither retraining uncertainty nor a multiplicity adjustment; differences in whether separate intervals cross zero do not establish a between-group interaction. Grouped AUCs are not an additive partition of overall AUC because cross-group positive–negative pairs remain.

A qualified case for cumulative and adaptive controls. On RankMixer–KuaiRand, the equal-active-parameter local control obtains 0.8546700 macro AUC, compared with 0.8615015 for full RISE, a difference of +0.0068315. The local construction also uses a fixed zero centering gauge; without a successful matching control-state audit,

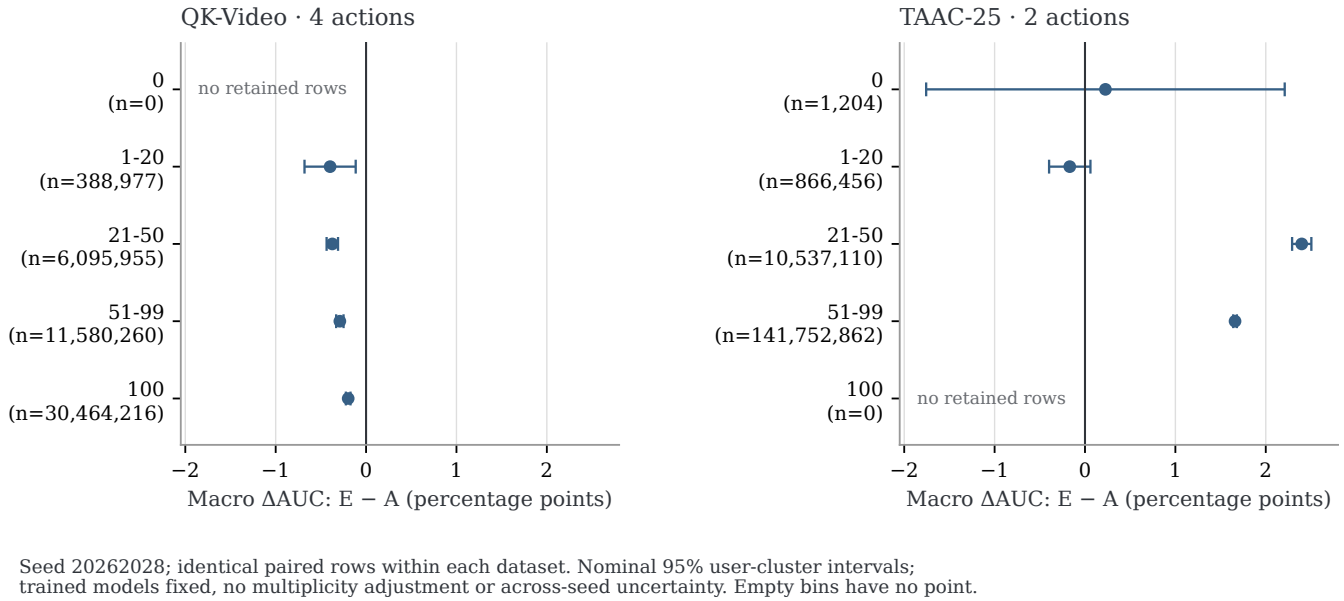


Figure 4: Full RISE minus independently trained native HiFormer by effective history length. Differences are percentage points of label-macro AUC; QK-Video and TAAC-25 have four and two actions, respectively. Error bars are nominal 95% user-cluster intervals for fixed models, not across-seed intervals.

this comparison cannot attribute every difference solely to accumulation under active clipping. The learned input-independent cumulative control instead obtains 0.8617022, slightly above full RISE in point estimate. On HyFormer–MerRec, the same input-independent principle obtains 0.7907552, below full RISE at 0.8008724 but above native A at 0.7856338. Thus, these experiments do not support a universal necessity claim for input-dependent controls. The fixed-control comparisons also change active parameter count and initialization; their single-seed point differences establish neither statistical equivalence nor a capacity-controlled causal effect of adaptivity.

Compiled selection observations. On a fixed diagnostic sample of 512 test interactions from 455 users, instrumented and uninstrumented compiled predictions are exactly identical for both checkpoints and all five actions. We reconstruct pre-dropout selection from actual captured queries, keys, and auxiliary corrections while retaining native padding semantics. Figure 5 shows equal-example mean total variation distance and sample percentiles at six sites. The full model's first sequence site has mean TV 0.638, compared with 0.188 for Content-only B; the first decoding site instead has mean TV 0.050 versus 0.089. These are within-model conditional changes, not native-backbone performance effects or a causal decomposition between separately trained models. Sample percentile whiskers are not confidence intervals. The sampling window is not population-representative. No successful observation of fused internal AM/PA states is claimed; unvalidated eager-state reconstructions are excluded from the interpretation.

Measured cost and attention backend. We benchmark native A, Content-only B, and full RISE E on one H100 using the same cached real inputs at batches 64, 256, and 512. Each variant runs in a fresh process with compiled BF16 inference, 1,000 warm-up forwards, and ten blocks of 100 timed forwards. Both execution orders, A–B–E and E–B–A, are retained. Across the six batch/order rounds, E/A forward-latency ratios range from 1.27 to 1.58. Profiling identifies a Flash SDPA path for native A and an efficient-attention path for B/E with explicit additive score bias. This backend difference is part of the measured implementation cost, not evidence that every mathematically equivalent implementation has the same overhead. At batch 512, peak allocated memory above the resident baseline is 384.53 MiB for A and 731.17 MiB for E; these quantities are not total model memory. Figure 6 shows both orders and within-round block IQRs, without selecting the faster round. The benchmark excludes loading, compilation, input transfer, and the data pipeline; it is neither serving tail latency nor end-to-end throughput. Low auxiliary dimension therefore does not by itself establish negligible realized inference cost.

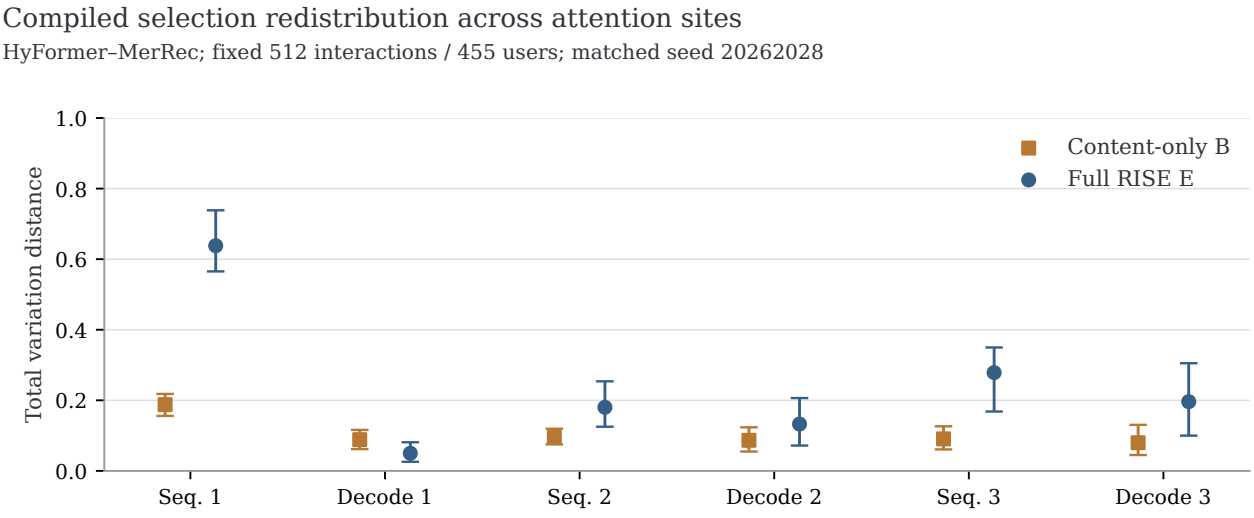


Figure 5: Selection redistribution in the compiled HyFormer–MerRec checkpoints. B retains auxiliary content projections and is not native A. Dots are means over the fixed diagnostic interactions; whiskers are their 10th–90th percentiles, not inferential intervals. One empty-history example has no valid sequence query and is excluded from sequence-site means.

Table 7: Additional QK-Video placement runs, all at seed 20262028. Native A is the first numerical column. No across-seed intervals are estimated.

Model	Placement	Native A	Full E	Selective
Zenith	history_only	0.9346347	0.9352205	0.9357857
HyFormer	sequence_only	0.9308580	0.9313080	0.9309360
Zenith	token_only	0.9346347	0.9352205	0.9338000
HyFormer	query_only	0.9308580	0.9313080	0.9312213

Data and inferential boundaries. The new comparisons remain single-seed, fixed-budget offline experiments. History length, activity, frequency, and label difficulty are correlated; descriptive strata do not identify their independent effects. For MerRec, the processed historical action encoding can contain later-attributed outcomes, while original action-event timestamps were unavailable in the scoped audit. Prediction-time availability and cross-split attribution rates therefore remain unresolved. These comparisons characterize the supplied offline encoding; they do not establish a leakage rate or certify online suitability.

Cross-backbone interpretation. On Zenith, history-only insertion improves macro AUC over native A, whereas token-only insertion reduces it. Full RISE remains above native A and has lower macro Logloss than history-only (0.07438775 versus 0.07440950), so the AUC-best placement is not uniformly best across metrics. On HyFormer, full RISE has the highest macro AUC, but its margin over query-only is just 0.00008675; follow and share remain below native A under full, sequence-only, and query-only insertion. HyFormer's sequence encoder and query-to-history decoder are distinct sequence-processing stages, not field-token ablation counterparts. These one-seed comparisons support architecture- and metric-dependent placement choices, without establishing seed-stable superiority for a universal insertion rule.

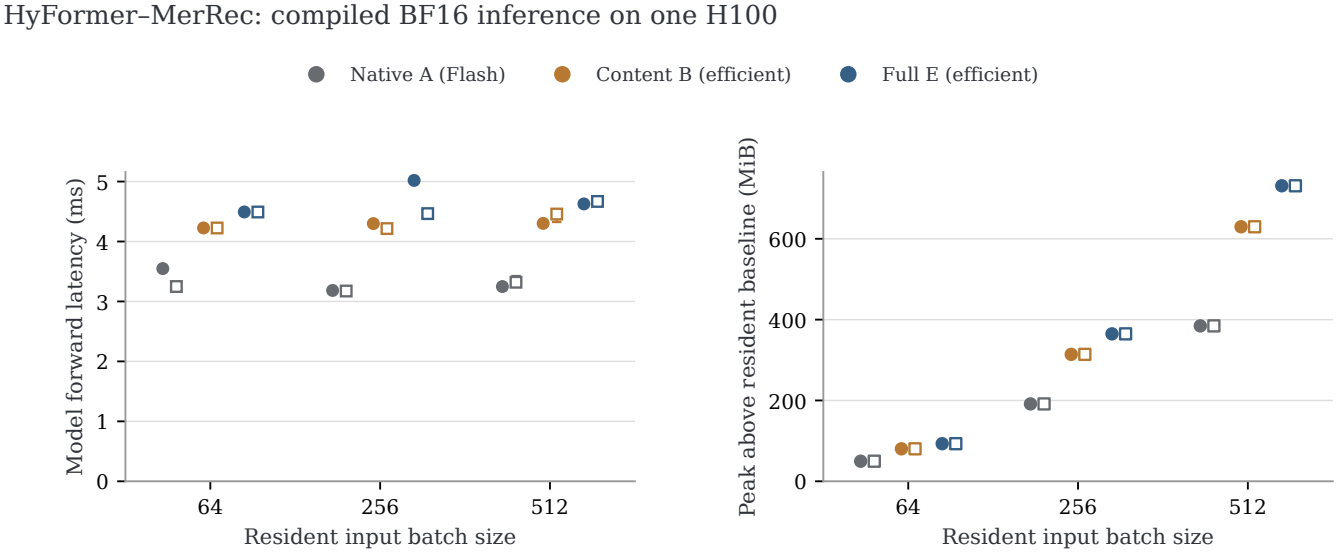
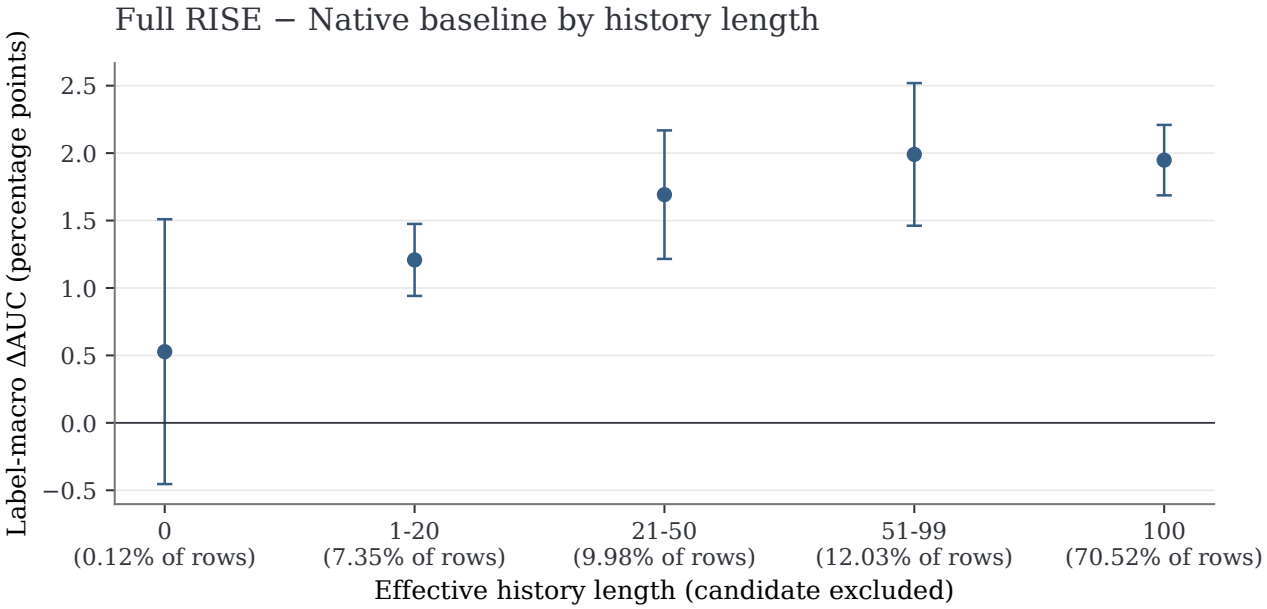


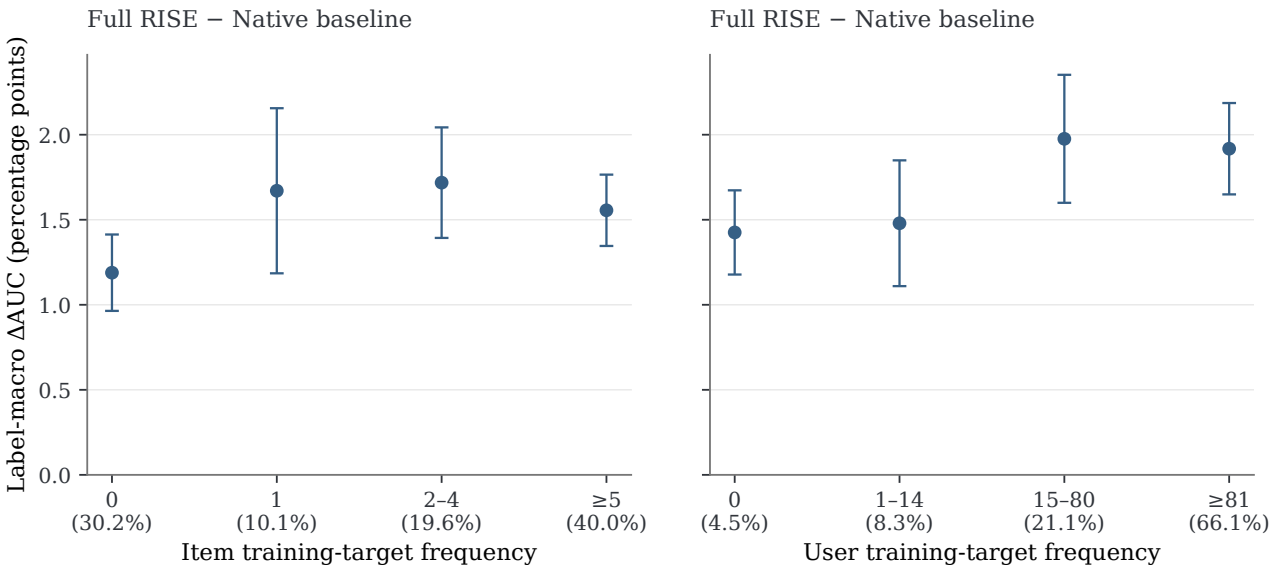
Figure 6: GPU-resident model-forward cost on the same H100. Filled circles and open squares retain the forward and reverse execution orders. Latency whiskers show the IQR of ten block means, not a confidence interval. Memory is incremental peak allocated above the warmed resident baseline.

3 Additional native-control figures



HyFormer–MerRec, matched seed 20262028; identical 11,599,872 retained test rows.
Nominal 95% user-cluster intervals, trained models fixed; the empty-history interval crosses zero.

Figure 7: Full RISE versus the native HyFormer backbone on MerRec by effective history length. Intervals condition on the trained models and use paired user-cluster influence functions.



HyFormer–MerRec; nominal 95% user-cluster intervals. Parentheses: share of retained test rows.
Cuts use positive-frequency training entities; 0 means unseen as a training target, not a strict cold-start claim.

Figure 8: Native-controlled MerRec effects by training-target frequency. Frequency zero is not proof of strict cold start.